

scphylo-tools: A Python toolkit for single-cell tumor phylogenetic analysis

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Summary

scphylo-tools is a Python library designed to unify single-cell tumor phylogeny inference methods. It addresses the lack of standardization in the field by providing a cohesive interface for data processing, tree reconstruction, visualization, and benchmarking. By streamlining these tasks, scphylo-tools empowers researchers with limited programming expertise to easily install and utilize complex inference methods, accelerates algorithm development, and ensures reproducible benchmarking of tumor phylogeny reconstruction methods.

Statement of Need

Cancer is a dynamic evolutionary process driven by the acquisition of somatic mutations and the competitive selection of clonal populations. Single-Cell Sequencing (SCS) technologies have enabled the profiling of genomic alterations at cellular resolution, offering unprecedented insights into intratumor heterogeneity (Navin et al., 2011; Wang et al., 2014). Elucidating the phylogenetic relationships between tumor cells is essential for understanding metastasis (Leung et al., 2017; Roper et al., 2020), drug resistance (Gopalan et al., 2021; Gruen et al., 2023; Kim & Simon, 2014), and clonal dynamics (Hirsch et al., 2025; Laks et al., 2019; Liu et al., 2025).

Reconstructing the evolutionary history of a tumor from SCS data presents unique computational challenges. These datasets are plagued by high noise levels, including Allele Drop-Out (ADO), false positives, missing data, and doublet artifacts (Malikić et al., 2021; Rashidi Mehrabadi, 2022). Consequently, a diverse array of computational tools has been developed to address these challenges, including stochastic methods like SCITE (Jahn et al., 2016), infSCITE (Kuipers et al., 2017), OncoNEM (Ross & Markowitz, 2016), SiFit (Zafar et al., 2017), and SiCloneFit (Zafar et al., 2019); combinatorial approaches such as PhISCS (Malikic et al., 2019), PhISCS-BnB (Sadeqi Azer et al., 2020), SPhyR (El-Kebir, 2018), ScisTree (Wu, 2019), gpps (Ciccolella, Soto Gomez, et al., 2020), and SAS (Ciccolella, Ricketts, et al., 2020); and specialized methods including HUNTRESS (Kizilkale et al., 2022), SCIPhI (Singer et al., 2018), and Scestial (Foroughmand-Araabi et al., 2022).

However, the software landscape remains highly fragmented. Existing methods typically function as standalone binaries or scripts with inconsistent input/output formats, rendering comparative analysis difficult. Researchers attempting to utilize these tools face a laborious process of installation, data conversion, and script wrapping. Furthermore, integrating these tools into modern Python-based environments (e.g., alongside SCANPY (F. A. Wolf et al., 2018) or Biopython (Cock et al., 2009)) often requires custom development. scphylo-tools addresses these challenges by providing a single, cohesive Python API that wraps independently developed standalone applications, including Trisicell (Mehrabadi et al., 2021), SCITE, PhISCS, and HUNTRESS.

41 The target audience includes computational biologists, bioinformaticians, and cancer researchers
42 who need to:

- 43 ▪ Compare multiple phylogeny inference methods on the same dataset
- 44 ▪ Benchmark new algorithms against established methods
- 45 ▪ Integrate tumor phylogeny analysis into existing Python workflows
- 46 ▪ Access curated single-cell cancer datasets for research

47 State of the Field

48 Several tools exist for tumor phylogeny inference, but none provide the unified framework
49 that scphylo-tools offers. Individual tools like SCITE, PhISCS, and HUNTRESS each
50 require separate installation, different input formats, and produce outputs that are not directly
51 comparable. Workflow frameworks like Snakemake or Nextflow could orchestrate these tools, but
52 would require significant custom development for format conversion and result standardization.

53 scphylo-tools distinguishes itself by:

- 54 1. **Unified Interface:** A consistent Python API across all wrapped methods, eliminating
55 format conversion overhead
- 56 2. **Comprehensive Metrics:** Built-in implementation of specialized tumor tree comparison
57 metrics (MLTD (Karpov et al., 2019), CASet/DISC (DiNardo et al., 2019), MP3
58 (Ciccolella, Bernardini, et al., 2020))
- 59 3. **Curated Datasets:** Direct access to published SCS datasets from landmark cancer
60 evolution studies (Gawad et al., 2014; Hou et al., 2012; Leung et al., 2017; Morita et al.,
61 2020; Wang et al., 2014; Y. Wolf et al., 2019)
- 62 4. **Ecosystem Integration:** Seamless integration with the Python scientific stack (NumPy,
63 NetworkX, Matplotlib)

64 Software Design

65 scphylo-tools follows a modular architecture inspired by SCANPY. Modules mirror the
66 analysis workflow: `io` (genotype and tree I/O, VCF export), `pp` (filtering of binary and read-
67 count matrices, node collapsing, subsampling), `tl` (solvers, tree-comparison scores, consensus
68 trees, partition functions, ancestral-state reconstruction), `pl` (tree rendering and genotype
69 heatmaps), `ul` (utilities), and `datasets`. A scphylo console script exposes the same operations
70 to shell users and workflow managers.

71 A single data model spans the package: genotype matrices are pandas DataFrames encoding
72 presence, absence, and missing entries; read counts are AnnData objects; trees are NetworkX
73 DiGraph objects. Over twenty solvers share the call shape `scp.tl.<method>(df_input, ...)`
74 and return a conflict-free matrix, so switching methods is a one-line change; serialization,
75 execution, parsing, and logging are internal. The ten tree-comparison metrics take the same
76 two arguments, reducing a benchmark to a short loop.

77 To remove the main installation barrier, performance-critical C++ codebases (SCITE, ScisTree,
78 MLTD, and the partition-function sampler) are vendored, compiled as Cython extensions, and
79 shipped as prebuilt wheels. Non-redistributable tools stay optional, located by one resolution
80 layer (explicit path, `SCPHYLO_TOOLS_DIR`, `scphylo.settings.tools_dir`, `PATH`) raising typed
81 exceptions that name the missing artifact and how to supply it. Heavy optional dependencies
82 (Gurobi, rpy2, graph-tool, Graphviz) load lazily.

83 Adding a method requires one module under `tl/solver` and no changes elsewhere, as with
84 the existing wrappers. For validation, `datasets` retrieves published SCS datasets by identifier
85 and the simulation engine generates ground-truth genotypes under configurable false-positive,
86 false-negative, missing-value, and doublet noise.

Research Impact Statement

scphylo-tools has enabled published research in three ways.

First, it provided the interface through which Trisicell and ScisTree were applied to build tumor phylogenies. These reconstructions underpinned the analysis of clonal sublines that uncovered heterogeneity-driven immunotherapy resistance in melanoma (Gruen et al., 2023), the comparison of genomic and epigenomic evolutionary trajectories in long-read sequencing of single cell-derived subclones (Liu et al., 2025), the stochastic modeling that quantified the non-genomic contribution to subclone evolution (Hirsch et al., 2025), and the recovery of metastatic migration histories from single-cell methylation sequencing (Liu et al., 2023).

Second, it supported the identification of expressed mutation profiles in single cells, which revealed subclonal expansion patterns and the therapeutic impact of intratumor heterogeneity (Mehrabadi et al., 2021).

Third, its uniform solver interface, simulation engine, and tree-comparison metrics were used to benchmark competing inference methods on common inputs during the evaluation of HUNTRESS (Kizilkale et al., 2022).

The package therefore serves as both the analysis layer for biological studies and the comparison harness for methods development. It is distributed on PyPI with source on GitHub and documentation on Read the Docs.

AI Usage Disclosure

No generative AI tools were used in the development of the scphylo-tools software, the writing of this manuscript, or the preparation of supporting documentation.

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